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Corresponding author(s): George M. Ibrahim

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Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed	
☐	☒	The exact sample size (n) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒	A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒	The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☐	☒	A description of all covariates tested
☐	☒	A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒	A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒	For null hypothesis testing, the test statistic (e.g. F, t, r) with confidence intervals, effect sizes, degrees of freedom and P value noted *Give P values as exact values whenever suitable.*
☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☐	☒	For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒	Estimates of effect sizes (e.g. Cohen's d, Pearson's r), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Research Electronic Data Capture (REDCap) v15.2.24 (Vanderbilt University) - Used for clinical data collection across centers.
Data analysis	Clinical Data Analysis: Python v3.12, Pandas v2.2, Numpy v1.24, Matplotlib v3.7, Seaborn v0.13, Statsmodels v0.14, UMAP-learn 0.5.7. Plotly 6.0.1, Scipy v 1.1.3 Deep Learning Modelling: Python v3.12, PyTorch v2.0.1, MONAI 1v.4.0, Scikit-learn v1.4.2, PyTorch Lighting v2.5.0, Antspyx v0.5.4, Nibabel v5.3.2, Torchvision v0.21, Torchio v0.20.4, Wandb v0.19.5, LPIPS v0.1.4, SHAP v0.47.1, Scipy v 1.1.3, intensity-normalization v2.2.4 (https://github.com/jcreinhold/intensity-normalization) Diffusion Imaging Procesing Python v3.12, Pandas v2.2, Numpy v1.24, Matplotlib v3.7, Seaborn v0.13, Statsmodels v0.14, Nibabel v5.3.2, TractoFlow 2.4.4 (https://github.com/scilus/tractoflow), Surface Enhanced Tractography (https://github.com/StongeEtienne/set-nf), COMMIT (https://github.com/daducci/COMMIT), DiPY v1.7, Scipy v 1.1.3 fMRI Processing: Python v3.12, Pandas v2.2, Numpy v1.24, Matplotlib v3.7, Seaborn v0.13, Statsmodels v0.14, Nibabel v5.3.2, AFNI 24.0.01 toolbox, Scipy v 1.1.3 Lesion Network Mapping: FreeSurfer v7.4.2, Python v3.12, Pandas v2.2, Numpy v1.24, Matplotlib v3.7, Seaborn v0.13, Statsmodels v0.14, Nibabel v5.3.2, Scipy v 1.1.3, DiPY v1.7 Suresh, H. Gmilab/VQVNS: V1.0.0. (Zenodo, 2026). doi:10.5281/ZENODO.18510266. (https://github.com/gmilab/VQVNS)

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

Deidentified data used in this study can be made available subject to the policies and procedures of the institution from which the data were collected. Data requests should be sent to the corresponding author. The raw data are protected and are not available due to data privacy laws.

Research involving human participants, their data, or biological material

Policy information about studies with human participants or human data. See also policy information about sex, gender (identity/presentation), and sexual orientation and race, ethnicity and racism.

Reporting on sex and gender	The sex for study participants is reported. 54% of subjects were male.
Reporting on race, ethnicity, or other socially relevant groupings	Race, ethnicity and other socially relevant groupings were not analyzed in this manuscript.
Population characteristics	The VNS cohort comprised 1,039 subjects, with a mean age of 10.6 years (SD = 4.7). Clinical data included continuous variables such as duration of epilepsy (mean = 7.42 years, SD = 4.38), and percent seizure reduction. Binary clinical variables included sex, presence of a genetic diagnosis (43%), developmental delay (71%), family history of epilepsy (28%), history of infantile spasms (14%), and congenital (25%) or acquired (19%) epilepsy etiology. Additionally, 13% of the cohort had undergone prior epilepsy surgery. EEG data indicated 53% had generalized and 48% had non-generalized seizure onset patterns, while 31% showed right-sided lateralization of interictal epileptiform discharges (IEDs).
Recruitment	Participants were enrolled under the CONNECTiVOS study protocol, which included both retrospective and prospective recruitment arms. For the retrospective cohort, data were collected from individuals who had previously undergone VNS implantation at participating institutions and met the inclusion criteria. For prospective recruitment, patients scheduled to undergo VNS were identified by their clinical care teams, who introduced the study to families. The study coordinator followed up with interested families to provide additional information, answer questions, and obtain informed consent from both the child and their parent or legal guardian. For children unable to provide consent, assent was obtained when possible and consent was secured from a substitute decision-maker.
Ethics oversight	This study was approved by the Hospital for Sick Children Research Ethics Board (REB number 1000061744).

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

[x] Life sciences [ ] Behavioural & social sciences [ ] Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	VNS Cohort: We included all available subjects with VNS across the 11 participating CONNECTiVOS sites (n = 1,039). For imaging-based modeling, we used the subset of 257 subjects with high-quality preoperative T1-weighted MRI and definitive seizure outcomes (80% or20% reduction). Pretraining Dataset: All 7,433 T1-weighted MRIs from publicly available and institutional datasets were used to support robust and diverse representation learning.
Data exclusions	VNS Cohort: Subjects were excluded if T1-weighted MRI was of poor quality (e.g., motion artifact, incomplete coverage), if there were significant natomical abnormalities (e.g., lissencephaly), or if outcome data were ambiguous or conflicting. Specifically, 47 subjects with abnormal anatomy were excluded from modeling but retained for qualitative analysis of reconstructions. Subjects with intermediate outcomes (20–80% seizure reduction) were excluded from the main classifier training but included in a secondary sensitivity analysis.
Replication	All model development and evaluations followed a training/validation/test split strategy. Predictive performance was confirmed via permutation testing (n = 1000). Results were robust to inclusion of subjects with intermediate outcomes. All preprocessing and analysis pipelines are publicly available to support replication.

Randomization	This was an observational study using registry and imaging data. Participants were not randomly assigned to groups. For modeling, data were randomly split into training, validation, and test sets. Stratified sampling was used to ensure balanced representation of responders and nonresponders in each split.
Blinding	Blinding was not applicable for data collection, as all data were obtained from retrospective/prospective clinical records and imaging repositories. Imaging-based outcome modeling was performed using deidentified data. During model development, the held-out test set was not accessed until final evaluation.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
[x]	[ ] Antibodies
[x]	[ ] Eukaryotic cell lines
[x]	[ ] Palaeontology and archaeology
[x]	[ ] Animals and other organisms
[ ]	[x] Clinical data
[x]	[ ] Dual use research of concern
[x]	[ ] Plants

Methods

n/a	Involved in the study
[x]	[ ] ChIP-seq
[x]	[ ] Flow cytometry
[ ]	[x] MRI-based neuroimaging

Clinical data

Policy information about clinical studies

All manuscripts should comply with the ICMJE guidelines for publication of clinical research and a completed CONSORT checklist must be included with all submissions.

Clinical trial registration	*Provide the trial registration number from ClinicalTrials.gov or an equivalent agency.*
Study protocol	Connectomic profiling and Vagus nerve stimulation Outcomes Study (CONNECTiVOS): a prospective observational protocol to identify biomarkers of seizure response in children and youth: https://pmc.ncbi.nlm.nih.gov/articles/PMC8995963/
Data collection	Clinical and imaging data were collected from 11 North American pediatric epilepsy centers as part of the CONNECTiVOS study. All patients underwent standard clinical evaluation for epilepsy surgery, including history, physical and neurological exams, video EEG, and T1-weighted MRI. For pretraining of the deep learning model, we compiled a diverse dataset of 7,433 T1-weighted MRIs from publicly available repositories and institutional archives. These included individuals with and without epilepsy, across a broad age range, scanner types, and acquisition protocols. All images were preprocessed using a harmonized pipeline prior to model training.
Outcomes	The primary outcome was seizure reduction at last follow-up, expressed as a percentage relative to baseline. Participants were classified as responders if seizure frequency was reduced by80% and as non-responders if20%, based on the bimodal distribution of outcomes. These thresholds minimized misclassification from subjective seizure reporting. Subjects with intermediate outcomes (20–80%) were excluded from the main prediction model but included in sensitivity analyses. Outcome data were obtained from clinical follow-up records, or parent reported surveys collected through the REDCap Registry.

Plants

Seed stocks	*Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.*
Novel plant genotypes	*Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.*
Authentication	*Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosiacism, off-target gene editing) were examined.*

Magnetic resonance imaging

Experimental design

Design type	Single T1w MRI acquired in the pre-operative period. Pretraining and normative data were acquired at a single timepoint.
Design specifications	*Specify the number of blocks, trials or experimental units per session and/or subject, and specify the length of each trial or block (if trials are blocked) and interval between trials.*
Behavioral performance measures	*State number and/or type of variables recorded (e.g. correct button press, response time) and what statistics were used to establish that the subjects were performing the task as expected (e.g. mean, range, and/or standard deviation across subjects).*

Acquisition

Imaging type(s)	We acquired only T1w images
Field strength	1.5T or 3T
Sequence & imaging parameters	The VNS cohort consisted of T1w images acquired with flip angle: 12.3° (8.0°–40.0°); TE: 3.7 ms (1.7–9.2 ms); TR: 7.9 ms (1.1–25.0 ms); voxel size: 0.7 × 0.7 × 0.7 mm. The SickKids internal normative T1w images were acquired at either 1.5T or 3T across various scanners (Signa HDxt, GE Healthcare; Achieva, Philips Healthcare; Magnetom Skyra, Siemens Healthineers) with a dedicated head-coil. Images were acquired either in the sagittal or the axial plane: TR/TE, 6-9/2-5 ms; flip angle: 10-15 degrees, mean voxel size, 1.0x1.0x1.5 mm.
Area of acquisition	Whole brain

Diffusion MRI ☒ Used ☐ Not used

Parameters *Specify # of directions, b-values, whether single shell or multi-shell, and if cardiac gating was used.*

Preprocessing

Preprocessing software	T1w: ANTs, FreeSurfer v7.4.2 dMRI (HCP): Preprocessed by dataset providers. Tractography performed with TractoFlow, SET, and COMMIT pipelines. fMRI (GSP): Preprocessed by dataset providers
Normalization	T1w: Rigid registration to 1mm isotropic MNI template (MNI152). Cortical thickness maps: Surface-based registration using FreeSurfer to fsaverage. fMRI (GSP): Functional data mapped to MNI152 (2 mm) space. dMRI (HCP): Structural alignment to MNI via cortical surface mapping and streamline projection.
Normalization template	T1w: 1 mm MNI152 Template fMRI: 2mm MNI152 Template Cortical surfaces: fsaverage spherical atlas. dMRI: Native HCP space enpoint map projected to 1mm MNI space. All analyses were projected to MNI152 1mm for cross modality comparisons
Noise and artifact removal	T1w: SynthStrip + SynthSeg for brain extraction; N4 bias field correction; visual QC of cortical surfaces. fMRI: Preprocessing completed by dataset providers. dMRI: Denoising, motion correction, and b0 normalization using TractoFlow.
Volume censoring	fMRI (GSP): Frames with FD > 0.2 mm or DVARS > 50 excluded. T1w & dMRI: Not applicable.

Statistical modeling & inference

Model type and settings	T1w (VNS cohort): Vector-quantized variational autoencoder (VQ-VAE) for latent encoding; linear SVM classifier for outcome prediction. Lesion network mapping: Vertex-wise cortical thickness z-maps, seeded to normative functional and structural data. fMRI (GSP): Seed-based functional connectivity maps; linear regression. dMRI (HCP): Streamline filtering and endpoint density mapping; voxel-wise t-tests between groups.
Effect(s) tested	Primary: Association between structural brain features and VNS response. Secondary: Concordance of model salience maps with neurotransmitter expression, and structural/functional network disconnection.

Specify type of analysis: ☒ Whole brain ☐ ROI-based ☐ Both

Statistic type for inference (See Eklund et al. 2016)	Clinical data correlations were modelled using ordinary least squared regression, and UMAP embedding. Clustering quality was evaluated using silhouette scores. Voxel-wise inference was used throughout for both structural and functional connectivity analyses. For lesion network mapping and fMRI-based connectivity, statistical comparisons were performed using voxel-wise t-tests. Deep learning model performance was evaluated using AUC, and permutation tested for significance (n = 1000).
Correction	Multiple comparisons for imaging data were corrected using spin permutation tests (n = 1,000) for spatial overlap analyses. For voxel-wise group comparisons in lesion network mapping, 3DClustSim (AFNI) was used: voxel-wise threshold $p < 0.05$; cluster-wise corrected = 0.05. For SHAP-salience correlation with neurotransmitter maps, Bonferroni correction was applied across receptor classes.

Models & analysis

n/a	Involved in the study
☐	☒ Functional and/or effective connectivity
☒	☐ Graph analysis
☐	☒ Multivariate modeling or predictive analysis

Functional and/or effective connectivity	Seed-based resting-state fMRI analyses were performed using the Brain Genomics Superstruct Project (GSP). Thresholded individual atrophy maps derived from cortical thickness z-scores ($z < -1.64$) were used as seeds. Pearson correlation was computed between each seed's time series and voxel-wise BOLD signals across the normative dataset. Correlation coefficients were Fisher Z-transformed, and voxelwise group comparisons were conducted using a general linear model (GLM) via FSL's randomise, with total atrophy volume as a covariate.
Multivariate modeling and predictive analysis	Independent variables: • Quantized latent representations of T1-weighted MRI derived from a vector-quantized variational autoencoder (VQ-VAE). Feature extraction: • The spatial mean of the 3D latent embedding was computed per subject to yield a 1D feature vector. • Features were filtered using two-sided t-tests ($p < 0.05$) to reduce dimensionality. Model: • Linear Support Vector Machine (SVM) with class weighting and regularization. • Trained on 77% of the VNS cohort (n = 199); validated on 10% (n = 25); tested on held-out 13% (n = 33). • AUC and confusion matrices were used for evaluation. • Empirical significance was determined by permutation testing (1,000 label shuffles).